DOD-C-24594(SH)
6 September 1979

METRIC

MILITARY SPECIFICATION

CELL, STORAGE, SILVER-ZINC ALKALINE TYPE

(FOR DEEP SUBMERGENCE APPLICATIONS) (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers silver-zinc alkaline storage cells used in a pressure compensated system subjected directly to sea pressures and temperatures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-626 - Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
QQ-B-639 - Brass, Naval: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-650 - Brazing Alloy, Copper, Copper-Zinc, and Copper-Phosphorus.
QQ-B-654 - Brazing Alloy, Silver.
QQ-C-502 - Copper Rods and Shapes; and Flat Products with Finished Edges (Flat Wire, Strips and Bars).
QQ-C-576 - Copper Flat Products with Slit, Slit and Edge - Rolled, Sheared, Sawed, or Machined Edges, (Plate, Bar, Sheet, and Strip).
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-S-365 - Silver Plating, Electrodeposited; General Requirements For.
PPP-B-566 - Box, Folding Paperboard.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-676 - Boxes, Setup.
PPP-C-300 - Chemicals, Liquid: Packaging and Packing of.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6140

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MILITARY (Continued)
DOD-P-15816 - Parts, Hard Rubber for Use with Submarine Storage Batteries (Metric).
DOD-P-15817 - Parts, Synthetic Rubber, for Use with Batteries or in Battery Compartment of Submarines (Metric).
MIL-I-45208 - Inspection System Requirements.
MIL-P-46120 - Plastic Molding and Extrusion Material, Polysulfone.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

CODE OF FEDERAL REGULATIONS DEPARTMENT OF TRANSPORTATION
Title 49 - Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, three cells shall be examined and tested as specified in 4.3 (see 6.3).

3.2 General requirements.

3.2.1 Design and construction. Each cell shall be rated at 750 ampere-hours (Ah) at the 5-hour rate and be capable of a series connection with one or more cells from the same production lot.

3.2.2 Installation and interface. Each cell shall be submerged in and be open to the pressure-compensating mineral oil filling the battery enclosure and serving as a buffer between the electrolyte and the seawater.

3.3 Electrical requirements.

3.3.1 Electrical rating. The rating of each cell based on an electrolyte temperature as low as 0°C ± 3°C at the start of the discharge and a simultaneous constant pressure as high as 62 megapascals (MPa) shall be not less than that shown in table I.

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TABLE I. Electrical rating.

Rate (amperes (A))	Time (hours (h))	Capacity (Ah)	Minimum average 1/ Voltage per cell (volts (V))	Minimum cell voltage (V)
240	3	720	1.40	1.25
150	5	750	1.45	1.34
95	8	760	1.50	1.36

1/ The "average voltage" is defined as the average of voltage recordings taken at equal intervals, starting with the first recording (taken between 1/2 minute and 1 minute after applying a constant rate) and including the last recording. The interval of recording shall be not greater than h/5 hours where h is the time in hours of the rated capacity of the test. The elapsed time at the last recording shall be not greater than the rated time.

3.3.2 Life. The cell, when installed, shall be capable of delivering a total of 40,000 Ah of discharge over a period of 18 months. The capacity shall be at least 70 percent of the rated capacity shown in table I for the first 26,000 Ah or 12 months of service, whichever occurs first. Thereafter, the capacity shall be at least 50 percent until the end of the guaranteed life (see 6.4).

3.3.3 Charge.

3.3.3.1 Full charge definition. Full charge is defined as an Ah input at least-equal to the Ah output obtained in the prior discharge before reaching the minimum cell voltage shown in table I.

3.3.3.2 Charge acceptance. The cell shall be capable of accepting 90 percent of full charge as defined in 3.3.3.1 in 16 hours after a discharge at any of the rates and for the corresponding times shown in table I, using the charging rate selected by the manufacturer (up to a maximum of 100 amperes) and with a starting electrolyte temperature as low as 10°C ± 3°C, and an ambient pressure as high as 7,000 kilopascals (kPa).

3.3.4 Hydrogen evolution. Hydrogen evolution from any charged cell shall not exceed 20 milliliters (mL) per hour average over a 24-hour period when the cell is maintained on open circuit at 25°C ± 3°C.

3.4 Physical requirements.

3.4.1 Cell mass. The mass of a filled and formed cell without compensating oil under the cover shall not exceed 14 kilograms (kg).

3.4.2 Cell jar. Cell jars shall be molded out of polysulfone in accordance with MIL-P-46120 using the Government-owned mold (see 3.8). Typical configuration is shown on figure 1. Cell jars shall be supported by neoprene pads 3 millimeters (mm) thick with 6 mm clearance around all edges.

3.4.3 Cell cover. Cell covers shall be molded out of the same material as the jar material using the Government-owned mold (see 3.8) specified in the contract or order (see 6.2.1). Typical configuration is shown on figure 2.

3.4.4 Cover-to-jar sealing. The cover-to-jar sealing shall be epoxy or cement.

3.4.5 Electrode pack.

3.4.5.1 Grids. The grids shall be in the form of expanded metal or perforated sheet metal or sheet silver metal. Positive active material shall be hot forged to each side of the positive grid. The equivalent thickness of any grid (positive or negative) when reduced to a solid sheet of same mass and same apparent electrode area shall be at least 0.05 mm.

3.4.5.2 Tabs. Silver shall be used for the positive and negative electrode tabs attached to the terminal base. The tab area below the top of the grid shall be at least 90 percent hot-forge bonded to the grid.

3.4.5.3 Silver active material. Silver active material amount shall be at least 3.35 grams per Ah, based on the 760 Ah capacity.

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3.4.5.4 Electrode construction. The positive electrodes and negative grids shall have all corners rounded to a radius of at least 6 mm. Edges of the positive and negative electrodes shall be smooth.

3.4.5.5 Separator system. The separator system shall be as follows:

(a) The separator system shall consist, from positive electrode to negative electrode, of a positive absorbent material layer, cellophane-type main separator layers, and a negative absorbent material layer.
(b) The positive and negative absorbent material shall be at least 0.12 to 0.17 mm-thick. The material shall be a layer of non-woven polyamide material from Pellon Corporation, or equal. The top shall be 5 to 6 mm over the top of the electrodes.
(c) A 0.25 nominal thickness layer of fuel cell grade asbestos absorbent material shall be included in the center of the positive electrode cellophane type main separator layers.
(d) The main separator shall be Dupont Cellophane PUDO-193 (battery grade as is or treated) in combination with Union Carbide Fibrous Sausage Casing (battery grade), or Yardney Electric Corporation C-19, or equal. The overall wet thickness of the combined layers of this material, when swollen in the specified electrolyte, shall be at least 0.45 mm. The main separator height over the top of the electrodes shall be as high as possible without contacting the terminal assembly.
(e) The main separator assembly method shall consist of bagging or wrapping the positive and negative electrodes separately. If bagging or wrapping of a pair of electrodes is used, it shall be done on the positive electrode. If wrapping is used, the overlap of the last layer shall be at least 15 mm.
(f) Epoxy or cement sealing of the bottom of separator shall not be used.

3.4.6 Terminals. Current carrying terminal parts shall be made of solid copper or copper alloys in accordance with QQ-B-613, QQ-B-626, QQ-B-637, QQ-B-639, or QQ-C-576. Terminal parts shall be silver-plated in accordance with QQ-S-365, type III, grade B, with a nickel undercoat in accordance with QQ-N-290, class 1, grade F, form SB. Assembly of the terminals shall be by mechanical attachment or brazing. Soldering shall not be used. Brazing shall be in accordance with MIL-B-7883, grade B, using brazing alloys in accordance with QQ-B-650 or QQ-B-654. Terminal posts shall be symmetrically placed with respect to an axis through the center of the cell top. One terminal post shall be provided for each polarity and shall be a part of a single, solid, silver-plated copper strap to which the plate tabs will be attached. Positive and negative terminals shall be identified.

3.4.7 Rubber parts. Soft rubber parts shall conform to DOD-P-15817. Hard rubber parts shall conform to DOD-P-15816. The material of O-rings, gaskets, and seals shall be neoprene or ethylene-propylene rubber, Hypalon or equal, except when used in the electrolyte entrainment eliminators (see 3.4.8.2), it shall be Hypalon.

3.4.8 Electrolyte. The electrolyte used for filling the cells shall be an aqueous KOH solution of 45 percent $\pm$ 1 percent KOH by mass with no additives.

3.4.8.1 Level. The manufacturer shall provide a means by which the electrolyte level may be determined for the purpose of electrolyte adjustment. At no time during operation shall the electrolyte level variation allow contact between compensating oil and electrodes.

3.4.8.2 Entrainment eliminators. Each cell shall be fitted with an electrolyte entrainment eliminator which shall be attached to the vent hole in the cell cover or part of the cover. The eliminator shall prevent carry-over of electrolyte from the cell to the oil-filled battery enclosure under the conditions of battery usage and operation. Baffles may be used under the vent hole to assist in entrainment elimination.

3.4.9 Firewalls. Individual cells shall have internal firewalls constructed of two or more layers of 0.25 mm nominal thickness fuel cell grade asbestos from Johns-Manville Corporation, enclosed in a polyethelene, heat-sealed bag, or equal.

3.4.10 Gas-tightness. Each cell shall be capable of withstanding 35 kPa differential air pressure for 5 minutes without loss of pressure.

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3.4.11 Cell identification.

3.4.11.1 Cell markers. A cell marker shall be attached to the top of each cell. The markers shall be consecutively numbered with each number preceded by a group number. For example, 01-10 indicates cell number 10 of group number 01. The group numbers and quantity of cells per group will be assigned by the Government (see 6.2.1).

3.4.11.2 Serial numbers. The manufacturer's serial numbers shall be visible from the top of the installed cells. The serial number shall consist of six digits, the first two representing the year of manufacture and the last four the manufacturer's serially assigned number.

3.4.12 Mercury. Mercury, or a mercury substitute as specified in the contract or order (see 6.2.1), shall be included in the negative active material.

3.4.12.1 Content. The amount of mercury (when converted to mercuric oxide) in the negative active material shall not exceed 2 percent of the total negative active material (when totally converted to zinc oxide).

3.4.12.2 Labeling and marking. The service manual and shipping instructions shall refer to the content, amount, and form of the mercury compound used in each cell. No cell label is required.

3.5 Environmental requirements.

3.5.1 Temperature. The cell shall operate submerged in seawater for extended periods, at temperatures from minus 2°C to 32°C. The cell shall also be suitable for exposure to air at temperatures from minus 30°C to 70°C.

3.5.2 Pressure. The cell shall be capable of spending its entire life with a compensating oil filling the top part of the cell. The cell shall withstand in this condition a compensating pressure as high as 62 MPa when submerged in seawater.

3.5.3 Angle of operation. The cell shall withstand 45 degree angles of operation continuously without physical or electrical damage. Electrolyte shall not spill nor oil come in contact with electrodes when the cells are tilted 45 degrees in any direction.

3.6 Cells, repair parts, and accessories.

3.6.1 Cells. The total number of cells to be delivered shall comply with the following:

(a) The specified (see 6.2.1) quantity of cells shall be shipped as specified (see 6.2.1).
(b) The specified (see 6.2.1) quantity of spare cells, for possible replacement of failed cells, shall be kept dry and shipped as specified (see 6.2.1).

3.6.2 Repair parts. Repair parts are all parts needed for repair, replacement, maintenance, and installation. Repair parts shall be provided as specified (see 6.2.1) and shall include the following for each group of cells:

(a) Fifteen entrainment eliminators with O-rings and netting.
(b) Requisite electrolyte for all dry spare cell filling (to be provided in separate polyethylene carboys of 3 to 4 liters).
(c) Two electrolyte level indicators.
(d) Two electrolyte adjustment devices.
(e) Eight cell markers (blank).

3.6.3 Accessories. The following accessories (parts for one cell) shall be provided as specified (see 6.2.1):

(a) Jar.
(b) Cover.

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3.7 Engineering services. Engineering services of the contractor's engineer shall be provided when specified (see 6.2.1) in order to:

(a) Assist the shipyard in preparing cells for installation and installing them.
(b) Assist the ship's force in becoming familiar with cell operation.
(c) Take corrective action in case of unexpected problems.
(d) Examine the cell condition when requested.
(e) Participate in technical meetings with NAVSEA or the ship's force with regard to cell operation and in locations other than the manufacturer's plant. Design review meetings shall not be considered a provision of engineering services.

A maximum number of mandays allowed for the entire contract or life of one set of batteries shall be as specified (see 6.2.1). Field engineering travel reports shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.8 Government-furnished items. The Government will make available the following items:

(a) Government-owned molds:
(1) Cell jar.
(2) Cell cover.
(b) Government-furnished material:
(1) Silver (see 3.4.5).

3.9 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 3.9.1 through 3.12.

3.9.1 Drawings. The contractor shall prepare engineering drawings (includes documents referenced therein) in accordance with the data ordering document included in the contract or order (see 6.2.2) for review by the Government as described in 3.9.1.1 and 3.9.1.2.

3.9.1.1 Nonproprietary documents. Nonproprietary documents shall be submitted to the Government for review and shall become the property of the Government. The nonproprietary documents shall include but not be limited to the following:

(a) Engineering drawings for each item specified in 3.4.
(b) Assembly drawing indicating all external dimensions and configuration of cell and hardware.
(c) A detailed drawing with dimensions (including tolerances) of the space below the cell cover from the bottom of the cover to the top of the electrodes and separators.

3.9.1.2 Proprietary documents. All proprietary engineering drawings (includes documents referenced therein) shall be accessible to the Government for review at any time and, when submitted in support of 3.9.1.1, shall be clearly marked as containing proprietary information. A data list identifying all proprietary documents shall be submitted.

3.10 Service manual. The contractor shall prepare a service manual in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.11 Milestone plan. The contractor shall provide and maintain a milestone plan and prepare a report in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.12 Engineering change proposals. The contractor shall prepare engineering change proposals in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.13 Design review. Design review meetings with Government representatives shall be held prior to the release of any drawings to contractors or production or to finalization of the service manual. At least three meetings (two at NAVSEA) are required: one prior to design to establish a design concepts agreement and provide any interface information, one to review drawings and design, and one to review the service manual.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.2.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Three cells shall be subjected to the first article inspection which shall consist of the examination specified in 4.5.2.1.1 and the tests specified in 4.5.2.2 and 4.5.3 through 4.5.3.10.

4.4 Quality conformance inspection. The entire production lot shall be subjected to the inspection of 4.5.2. The sample cells (see 4.4.1) shall be filled, formed, and subjected to the inspections specified in 4.5.3 through 4.5.3.10.

4.4.1 Sampling for quality conformance inspection. A sample of six cells shall be randomly selected from the production lot for quality conformance inspection (see 4.4).

4.5 Inspection methods.

4.5.1 Inspection conditions. The tests specified herein shall be started as soon as possible after selection of cells. The test cycles shall succeed one another as rapidly as temperature limitations permit. Except where otherwise specified, the temperature of the electrolyte at the start of each test discharge and charge shall be between 15°C and 32°C. After the cells are filled and formed in accordance with the manufacturer's instructions, they shall be charged and submitted to the tests specified in 4.5.3 through 4.5.3.10.

4.5.2 Inspection of all cells in the dry condition. The examination and test specified in 4.5.2.1 and 4.5.2.2 are applicable to parts and cells manufactured under the same contract, in the dry condition.

4.5.2.1 Examination.

4.5.2.1.1 First article examination. Parts used in the assembly of the cells shall be examined for conformance to the requirements of this specification and to the manufacturer's quality control requirements.

4.5.2.1.2 Quality conformance examination. Parts used in the assembly of the cells shall be examined on a sample basis in accordance with MIL-STD-105, inspection level II, acceptance quality level (AQL) equals 1.0 percent defective for conformance to the requirements of this specification and to the manufacturer's quality control requirements. Cells shall be examined for workmanship and for conformance to 3.4.2, 3.4.3, 3.4.4, 3.4.6, and 3.4.11. Any defective cell shall be rejected unless the correction proposed by the manufacturer is submitted as specified in 4.5.3.9, in which instance the cell shall be resubmitted.

4.5.2.2 Gas tightness. The cells shall be tested to determine conformance with 3.4.10.

4.5.3 Inspection of first article and sample cells in the wet condition.

4.5.3.1 Mass. The first article and sample cells shall be weighed. After formation, the mass shall meet the requirements specified in 3.4.1.

4.5.3.2 Individual cell tests.

4.5.3.2.1 Capacity and voltage at 26°C. Discharge each cell at 95 A until it reaches 1.36 volts. At this point in life, the cell shall give at least 836 Ah.

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4.5.3.3 Charge acceptance. Each cell shall be tested for charge acceptance to determine conformance with 3.3.3.2 following any discharge rate.

4.5.3.4 Hydrogen evolution. Each cell shall be tested for hydrogen evolution on open circuit stand to determine conformance with 3.3.4. The open circuit voltage period shall start after the end of a charge conducted in accordance with the manufacturer's service manual. Gas collection shall commence at the beginning of the open circuit period and quantities shall be recorded periodically until completion of the test.

4.5.3.5 Capacity and voltage determination. After the test specified in 4.5.3.2.1, each cell shall be cycled at room temperature as follows:

(a) Cycle 1. Each cell shall be charged as recommended in the service manual, then placed in an oil-filled vessel, pressurized to 62 MPa. Each cell shall then be discharged at 240 A to the corresponding minimum cell voltage.
(b) Cycle 2. Each cell shall be depressurized at the rate of 700 kPa per minute down to atmospheric pressure and then recharged. Each cell shall then be pressurized to 62 MPa. The cell shall then be discharged at 150 A to the corresponding minimum cell voltage. The cell shall be depressurized at the rate of 700 kPa per minute down to atmospheric pressure.
(c) Cycle 3. Each cell shall then be recharged at 100 A to the recommended cut-off voltage then immediately pressurized to 62 MPa then discharged at 200 A for 2 hours. The cell shall then be depressurized at the rate of 700 kPa per minute.

Data on voltage, current, and electrolyte termperature shall be monitored and recorded. The results of the capacity discharge tests at each rate shall be averaged and the estimated standard deviation calculated according to the following formula:

$$s = \left(\frac{\sum_{1}^{n} (x_i - \bar{x})^2}{n-1} \right)^{1/2}$$

Where:
s = Estimated standard deviation.
n = Number of sample cells.
$\bar{x}$ = Average of results of test.
x_i = Result of individual sample cell.

Capacity and voltage of each cell shall meet the requirements of 3.3.1 and the average percent of rated capacity for all cells shall be equal to or greater than 100 percent of rated capacity plus 3.1 s.

4.5.3.6 Angle of operation. Each cell, within 1 hour after ending any charge, shall be tilted in any direction 45 degrees from the vertical without spilling electrolyte. The cells shall be tilted, for information only, 60 degrees from the vertical or to an angle less than 60 degrees where spill occurs, to determine the maximum inclination capability of the manufacturer's design.

4.5.3.7 Electrolyte level indication. The method of determination of the actual electrolyte level shall be demonstrated on actual cells.

4.5.3.8 Electrolyte entrainment eliminator and cycling under pressure. The effectiveness of the electrolyte entrainment eliminator shall be demonstrated during the tests of 4.5.3.5. There shall be no visual evidence of electrolyte carry-over out of any cell during the entire sequence of tests.

4.5.3.9 Failure procedure. In case of failure, the tests of 4.5.3 may be repeated once. In case of subsequent failure, the lot represented by the sample cells which fail to meet individually the tests of 4.5.3 shall be rejected and may not be offered again for delivery unless a proposal for the correction of the deficiency and retest of new sample cells (replacing the failed cells) has been submitted as specified in 4.6.2.3.

4.5.3.10 Disposition of first article cells and sample cells. The first article cells and sample cells shall be shipped to a Government laboratory, as specified (see 6.2.1).

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4.6 Test procedures and reports. The contractor shall prepare test procedures and reports in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 4.6.1 through 4.6.2.5.

4.6.1 Test procedures.

4.6.1.1 First article and quality conformance procedures. The contractor shall prepare first article and quality conformance inspection (see 4.2) procedures in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.6.2 Test reports.

4.6.2.1 First article inspection report. The contractor shall prepare first article inspection (see 4.3) report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.6.2.2 Sample cell (wet condition) test report. The contractor shall prepare a sample cell (see 4.5.3) test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.6.2.3 Cell failure analysis. For each cell failed (see 4.5.3.9) in the manufacturer's plant, a full cell analysis shall be made and a report shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.6.2.4 Electrolyte analysis report. A report of analysis for impurity content and for potassium carbonate content shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2), prior to the use of electrolyte in any cell or battery.

4.6.2.5 Silver metal accountability report. A report on accounting of silver metal furnished by the Government and used in the cells delivered shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2). The report shall include the amount of silver received by the contractor, the amount per cell, broken down into active material and inactive material (grid and plating), and the amount used in the cells delivered.

4.7 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING (see 6.5)

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cells (see 3.6.1). Cells shall be individually unit protected method III in accordance with MIL-P-116 and packed as specified in 5.2.2.1.

5.1.1.2 Repair parts and accessories. Repair parts and accessories (see 3.6.2 and 3.6.3) shall be segregated and each item in the quantity required for use, except as specified herein, shall be unit protected method III in accordance with MIL-P-116 in individual paperboard folding or set-up cartons or fiberboard boxes conforming to PPP-B-566, PPP-B-676, or PPP-B-636 respectively, domestic grade, except as follows:

(a) Cell jars - One to a fiberboard box.
(b) Cell covers - Ten to a fiberboard box. Each cover shall be wrapped or interleaved with a resilient cushioning material.
(c) Electrolyte - Electrolyte shall be packaged in 3 to 4 liter plastic containers conforming to the level A requirements of PPP-C-300.
(d) Cell, complete with or without electrolyte - One to a fiberboard box. Cells with electrolyte shall require the addition of a plastic bag prior to placing in the fiberboard box to trap electrolyte in case of spillage during shipment.

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5.1.2 Level C.

5.1.2.1 Cells. Cells shall be packaged as specified in 5.1.1.1.

5.1.2.2 Repair parts and accessories. Repair parts and accessories shall be segregated and each item, in the quantity required for use, shall be packaged in a manner which will afford protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging methods may be used when such meets the requirements of this level.

5.1.2.3 Electrolyte. Electrolyte shall be packaged as specified in 5.1.1.2(c).

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. Cells, repair parts and accessories packaged as specified in 5.1, shall be packed as specified in 5.2.2 except the containers shall be of the overseas type or class.

5.2.2 Level B.

5.2.2.1 Cells. Cells packaged as specified in 5.1.1.1 and 5.1.1.2 shall be packed in a wood-cleated plywood container in accordance with PPP-B-601, domestic type, style optional and the following:

(a) Ten cells shall be packed per container.
(b) Containers shall accept either filled (wet) or empty (dry) cells.
(c) Container tops shall be secured by screws in lieu of nailing (see 5.2.2.1(g)).
(d) Cells shall be segregated within the container by separators or partitions.
(e) Containers with filled cells shall be modified by the addition of structural members to prevent container distortion.
(f) Containers shall have a minimum of two wood skids in accordance with the container specification.
(g) Containers shall be reusable.

5.2.2.2 Spare and test cells.

5.2.2.2.1 Spare cells. Cells packaged as specified in 5.1 shall be packed as specified in 5.2.2.1.

5.2.2.2.2 Test cells. Cells packaged as specified in 5.1 shall be packed in fiberboard boxes conforming to PPP-B-636, class domestic, or PPP-B-640, class 1. Boxes shall be closed in accordance with the applicable container specification or appendix thereto.

5.2.2.3 Repair parts and accessories (less electrolyte). Repair parts and accessories packaged as specified in 5.1 shall be packed in fiberboard boxes conforming to PPP-B-636, class domestic or PPP-B-640, class 1. Alternatively, wood containers conforming to PPP-B-601, domestic type; PPP-B-621, class 1; PPP-B-576, class 1; or PPP-B-585, class II may also be used. Container and style selection shall be at the option of the contractor. Container closure and reinforcing as applicable, shall be in accordance with the applicable container specification or appendix thereto.

5.2.2.4 Electrolyte. Electrolyte, packaged as specified, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, as specified in Department of Transportation Regulations (DOT), Title 49, section 173.260, subpart E, gross mass shall not exceed 30 kg. Alternatively, electrolyte packaged as specified shall be packed in wood containers specified in 5.2.2.3 with suitable padding and separators between the electrolyte carboys and inside faces of the wood containers.

5.2.2.5 Consolidated packing. Items specified in 5.2.2.3, when supplied with a complete battery, may be consolidated within a single shipping container provided the items are segregated by intermediate packing utilizing fiberboard boxes. The container(s) shall be provided with two index lists completely itemizing and identifying the container contents. One copy shall be placed inside the container in a location offering ease of accessibility and removal. One copy shall be attached to the exterior side or end of the container in a protected location which will prevent loss or damage to the list during shipment and storage. Lists shall be enclosed in a waterproof sealed bag or envelope.

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5.2.3 Level C.

5.2.3.1 Cells. Cells shall be packed as specified in 5.2.2.1 and 5.2.2.2.

5.2.3.2 Repair parts, accessories, and electrolyte. Repair parts, accessories, and electrolyte, packaged as specified, shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight and DOT rules and regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2.1) or herein, interior packages (unit and intermediate) and shipping containers shall be marked in accordance with MIL-P-116 and MIL-STD-129 as applicable to the levels of packaging and packing specified.

5.3.1 Special markings.

5.3.1.1 Cell containers. Cell containers shall be marked with the following:

(a) "DO NOT DESTROY - REUSABLE."
(b) "FRAGILE" and up-arrows (see MIL-STD-129).
(c) "STORE IN A COOL AREA" (preferably minus 7°C to 4°C) and other storage criteria deemed necessary by the contractor for guidance to the receiving and storing activity for filled and dry cells.
(d) Labelling as required by the Department of Transportation.
(e) Structural and handling markings such as center-of-balance and lift points.
(f) Asbestos (see 3.4.5.5).

5.4 Technical data (see 3.9, 4.6, and 4.7). When specified (see 6.2.1), preservation-packaging, packing, and marking drawings shall be submitted (see 6.2.2). Drawings are not required when previously submitted and reviewed. Drawings, service manuals, test procedures and reports furnished to the contracting activity shall be prepared for shipment in accordance with level C requirements (see 5.1.2 and 5.2.3).

6. NOTES

6.1 Intended use. Cells covered by this specification are intended for use in the deep submergence vehicles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Group numbers and quantity of cells per group (see 3.4.11.1).
(c) Mercury substitute, if required (see 3.4.12).
(d) Address for shipment of the cells (see 3.6.1).
(e) Address for shipment of the spare cells (see 3.6.1).
(f) Total number of cells to be delivered (see 3.6.1).
(g) Address for shipment of repair parts and quantity (see 3.6.2).
(h) Address for shipment of accessories, and quantity if accessories are required (see 3.6.3).
(i) If engineering services are required and the maximum number of mandays allowed (see 3.7).
(j) Quality assurance requirements (see 4.1.1).
(k) Address for shipment of first article, when required, and sample cells to a Government laboratory (see 4.5.3.10).
(l) Level of preservation-packaging and methods required (see 5.1).
(m) Level of packing and methods required (see 5.2).
(n) Special marking required (see 5.3).
(o) When preservation-packaging, packing, and marking drawings are required (see 5.4).

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6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.7	Report, contractor engineering and technical services (CETS)	UDI-E-23127	------
3.9.1 and 5.4	Drawings, engineering and associated lists	DI-E-7031	Level 2 Design activity designation - contractor Drawing number - contractor
3.10	Technical publications for advanced development programs	DI-M-6157	------
3.11	Plan, milestone (type II) for critical supply contracts	UDI-A-23015	------
3.11	Report, milestone (type II)	UDI-A-23018	------
3.12	Engineering change proposals (ECPs) and requests for deviations and waivers (long form)	DI-E-2037	------
4.6.1.1	Procedures, test	UDI-T-23732	------
4.6.2.1	First article inspection report	DI-T-4902	------
4.6.2.2, 4.6.2.3, 4.6.2.4, and 4.6.2.5	Reports, test	DI-T-2072	------

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Guarantee. The guarantee of the cell when series-connect as a battery shall be as specified in 6.4.1 through 6.4.4.

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6.4.1 Nothing in the following guarantee provision shall limit any of the rights afforded to the Government by the provisions of the inspection clause including those provisions relating to latent defects, fraud, or gross mistakes that amount to fraud. This guarantee applies notwithstanding inspection and acceptance by the Government or any provision of the contract concerning the conclusiveness thereof. The contractor guarantees the performance of the battery in terms of service operating conditions (months of service or total Ah output) when the battery is operated and maintained in accordance with the service manual. The battery shall be guaranteed for 18 months or to deliver 40,000 Ah, whichever occurs first, before the battery capacity drops to 50 percent of rated capacity as specified herein, when tested at any rate in table I and subject to the temperature limitations of this specification. The period of the guarantee shall begin:

(a) Date the cells are filled and formed in accordance with the manufacturers instructions, or
(b) 24 months after acceptance of the sample cell test report in accordance with 4.6.2.2, whichever occurs first.

6.4.2 In the event of a failure of the battery or at least five cells, as specified in 6.4.3 the contractor shall:

(a) When battery capacity drops below 70 percent of rated capacity as specified herein, when tested at any rate in table I and subject to the temperature limitations of this specification within the first 12 months or 26,000 Ah, whichever occurs first, the contractor shall replace the total battery, with a new unused battery manufactured in accordance with these specifications at no cost to the Government.
(b) After the period specified in (a) and prior to the end of the guarantee life of 18 months or 40,000 Ah, whichever occurs first the contractor shall indemnify the Government on a pro-rata basis (proportionately) for months of service not obtained or total Ah output not obtained, whichever is smaller. (If P being the purchase price of the battery, N the number of months not obtained A the total Ah output not obtained, the smaller of the two numbers $P\frac{N}{18}$ and $P\frac{Q}{40,000}$ shall be used.)

6.4.3 The battery shall be considered failed in the following cases:

(a) When grounds develop which cannot be corrected by the contractor to meet the requirements of the service manual. Corrective action by the contractor shall be at his own expense.
(b) When the battery minimum voltage specified in table I limits battery capacity to a percentage specified in 6.4.2 and cannot be restored by contractor corrective action which will be at his own expense.
(c) When five or more cells fail as defined below:

Any cell shall be considered failed in the following cases:

(a) The cell is internally shorted.
(b) The cell limits the battery capacity to a percentage specified in 6.4.2 and has to be removed in order to raise the overall battery capacity.
(c) The cell develops leaks through cracks, thus causing grounds in the system.

6.4.4 The same guarantee requirements of life, total Ah output, minimum capacity and indemnification specified in 6.4.1, 6.4.2 and 6.4.3 shall apply to each spare cell individually, whether wet or dry. Nothing in 6.4 diminishes the Government's rights as contained in any other provision of the contract, including but not limited to inspection.

6.5 The packaging and packing requirements of DOT are exempted for shipments of filled cells when such shipments comply with section 173.260(e), parts (1) through (4) of Title 49 DOT Regulations. Contracts or orders shall contain a statement requiring exclusive use of the carrier vehicle as follows:

"Special handling with exclusive use and through service shipment required for _____ (fill in applicable contract/order line item number). To be routed via specialized carriers employing heavy duty equipment. Motor freight vehicles shall be equipped with air ride suspension."

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6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 6140-N512)

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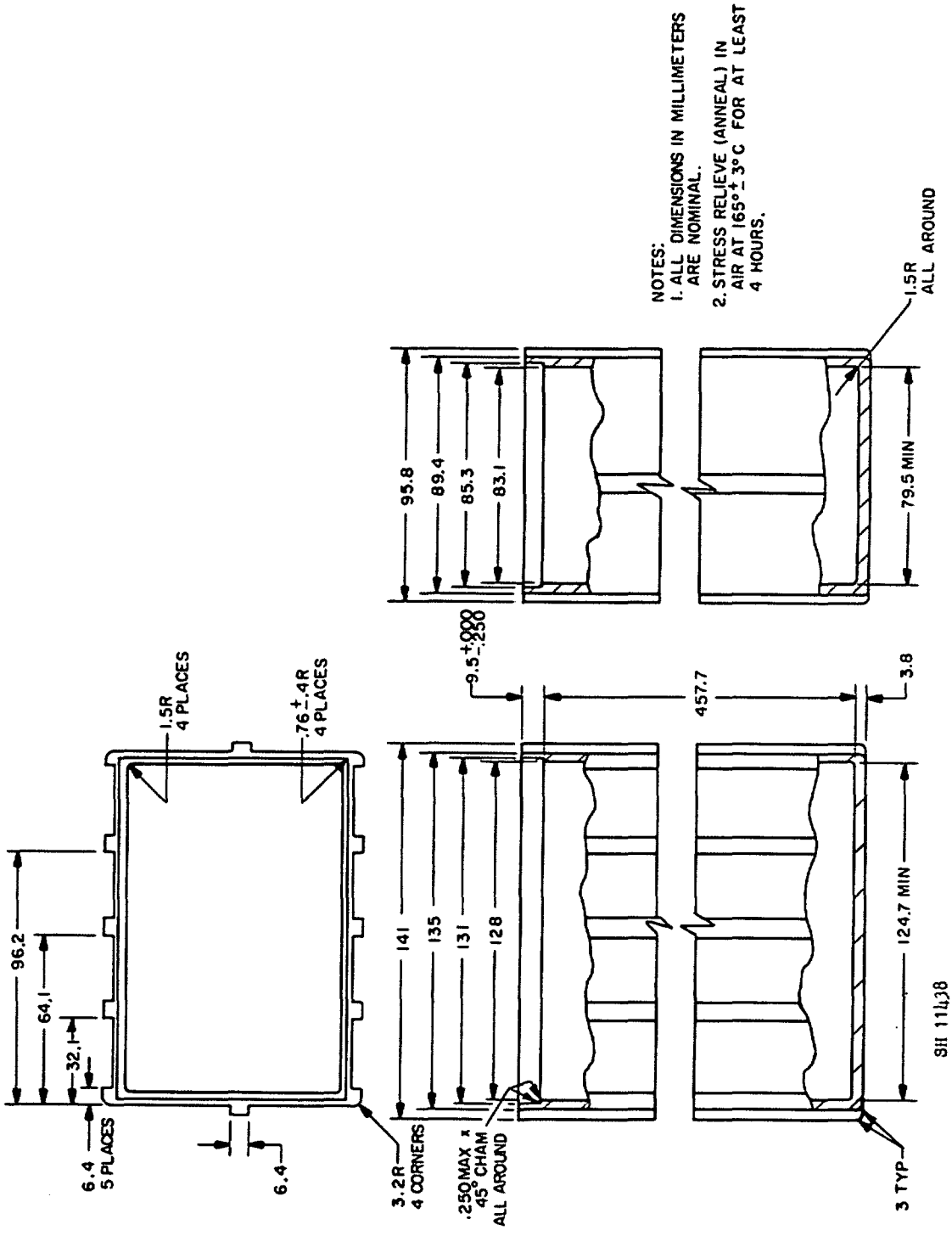
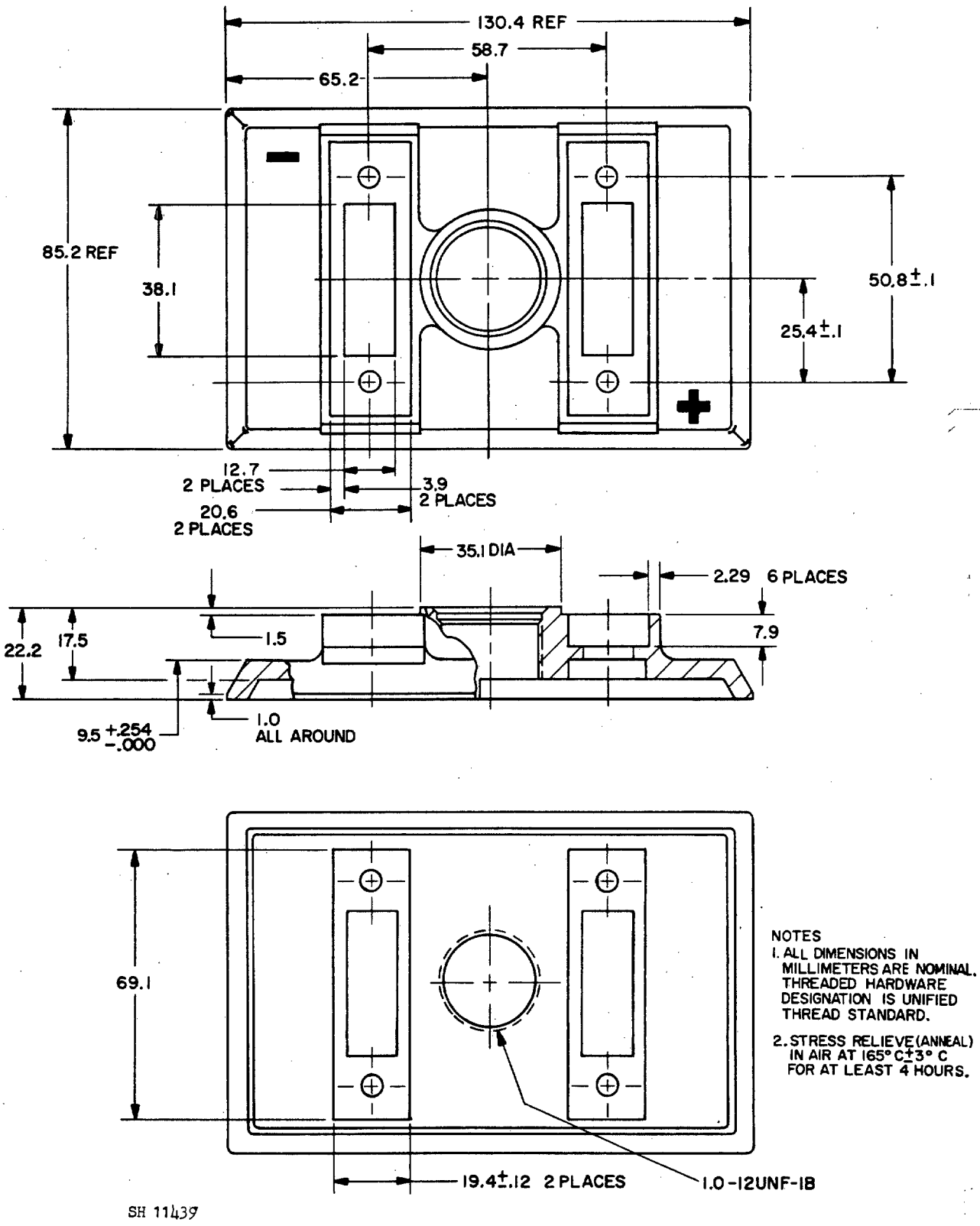


FIGURE 1. Cell jar.

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NOTES
1. ALL DIMENSIONS IN MILLIMETERS ARE NOMINAL. THREADED HARDWARE DESIGNATION IS UNIFIED THREAD STANDARD.
2. STRESS RELIEVE(ANNEAL) IN AIR AT 165°C±3° C FOR AT LEAST 4 HOURS.

FIGURE 2. Cell cover.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-24594(SH)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT† CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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